

YIFAN ZHU

Northwestern Polytechnical University | Software Engineering Undergraduate,
Junior Year

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Laurent-Zhu

Xi'an, Shaanxi



EDUCATION

B.Eng. in Software Engineering
Northwestern Polytechnical University
2023.09 – Present

- Academic Performance: GPA 3.813/4.1, ranked 11/270
- English Proficiency: CET-4 613; CET-6 587; CET-6 Speaking Test: Excellent
- Awarded the National Scholarship for Undergraduate Students for two consecutive years

RESEARCH EXPERIENCE

BrepLLM: Enabling Large Language Models to Understand Boundary Representations
ECCV 2026 Conference Submission, Fifth Author

Unified Multimodal Agent Framework for CadQuery-based 3D-CAD Generation, Understanding and Editing
Ongoing Research Project

- Participating in a research project on a multimodal agent framework for 3D-CAD generation, understanding, and editing, using executable Python CadQuery programs as the unified modeling representation. My work focuses on high-quality CadQuery data construction, unified SFT data format design, model fine-tuning, and CAD-kernel-based automated evaluation and repair.

INTERNSHIP EXPERIENCE

Shenzhen Kuakua Jingling Technology Co., Ltd.
Back-end Developer (Intern)
2025.07 – 2025.08

SELECTED HONORS AND AWARDS

International Level

- Honorable Mention, 2026 Mathematical Contest in Modeling (MCM) (2026.05)

National Level

- National Scholarship for Undergraduate Students (2024.12)
- National Scholarship for Undergraduate Students (2025.12)
- Second Prize, Special Track of the 19th "Challenge Cup" National College Students' Extracurricular Academic Science and Technology Works Competition (2024.11)

Provincial / Regional Level

- Third Prize, 16th National College Students Mathematics Competition, Non-Mathematics Category A (2024.12)
- Third Prize, Northwest Regional Contest, 18th National Undergraduate Software Innovation Competition – Software Design Innovation Contest (2025.04)
- Third Prize, Shaanxi Division, 2025 11th National College Students Statistical Modeling Competition (2025.05)
- Second Prize, Shaanxi Division, 2025 Contemporary Undergraduate Mathematical Contest in Modeling (2025.11)

PROJECT EXPERIENCE

Dual-modal UAV Urban Traffic Perception and Intelligent Analysis System
Undergraduate Innovation Training Project

- Developed a visible-light and infrared UAV traffic perception system using a customized YOLOv8-OBb fusion model on the DroneVehicle dataset.
- Built an Edge-Cloud-Dashboard architecture with WebSocket streaming, FastAPI services, and React-based real-time visualization.